

Non-Functional Requirements

Interview Questions — Clothing Store E-Commerce System

Database: Microsoft SQL Server

Use this questionnaire when interviewing the clothing store owner to gather non-functional requirements before designing the e-commerce system.

1. Performance & Response Time

Q1. How many concurrent users do you expect during normal hours vs. peak sales events (e.g., Eid, Black Friday)?

■ This drives SQL Server connection pool sizing and indexing strategy.

Q2. What is the maximum acceptable page load time for product listing and checkout pages?

■ Helps set performance benchmarks; typical target is under 3 seconds.

Q3. Do you expect a large product catalog? Roughly how many SKUs (stock keeping units) will be listed?

■ Affects SQL Server query optimization, indexing on Products table, and pagination.

Q4. Are there specific operations (e.g., search, order placement) that must complete within a defined time?

2. Scalability & Growth

Q5. Do you plan to expand to multiple branches or warehouses in the future?

■ Impacts whether the SQL Server schema needs multi-tenant or multi-location support.

Q6. How fast do you expect the product catalog and order volume to grow over the next 2–3 years?

Q7. Will the platform support multiple currencies or international shipping in the future?

3. Availability & Reliability

Q8. What are your expected business hours? Should the site be available 24/7 or only during certain hours?

■ Determines uptime SLA; e.g., 99.9% uptime = ~8.7 hrs downtime/year.

Q9. What is the maximum tolerable downtime per month (planned + unplanned)?

Q10. In case of a SQL Server failure, how quickly must the system recover? (Recovery Time Objective)

■ Guides SQL Server Always On, failover clustering, or simple backup restore strategy.

Q11. How much data loss is acceptable in a disaster scenario? (Recovery Point Objective)

■ Defines SQL Server backup frequency — full, differential, transaction log backups.

4. Security

Q12. What customer data will be stored (names, addresses, payment info)? Are you required to comply with any data privacy laws (e.g., GDPR, PCI-DSS)?

■ Affects SQL Server encryption (TDE), column-level encryption, and audit logging.

Q13. Will payment processing happen directly on your platform or via a third-party gateway (e.g., Stripe, JazzCash)?

■ If on-platform, strict PCI-DSS compliance applies to the SQL Server database.

Q14. Who within your organization should have access to the database? Are there different roles (admin, manager, staff)?

■ *Maps to SQL Server Role-Based Access Control (RBAC) and login permissions.*

Q15. Should all sensitive columns (e.g., passwords, card tokens) be encrypted at rest in the database?

■ *SQL Server supports Always Encrypted and Transparent Data Encryption (TDE).*

5. Maintainability & Manageability

Q16. Do you have an IT team to maintain the database and server, or will this be managed by an external vendor?

Q17. How often will products, prices, and promotions be updated? Who will manage these updates?

Q18. Do you need detailed audit logs of who changed what data and when?

■ *SQL Server supports CDC (Change Data Capture) and audit specifications for this.*

6. Backup & Data Recovery

Q19. How often should the database be backed up (daily, hourly)?

■ *SQL Server supports full, differential, and transaction log backups.*

Q20. Where should backups be stored — on-premises, cloud (Azure Blob), or both?

Q21. How long should order and customer data be retained before archiving or deletion?

■ *Affects SQL Server partitioning or archival table strategy.*

7. Usability & Accessibility

Q22. What devices will your customers primarily use — mobile, desktop, or both?

■ *Influences front-end decisions but also impacts query result set sizes from SQL Server.*

Q23. Do you need multi-language support on the storefront (e.g., English, Urdu)?

■ *May require a localization table in SQL Server.*

Q24. Should the admin panel be usable by non-technical staff with minimal training?

8. Compliance & Legal

Q25. Are you required to generate tax invoices or integrate with any government tax system (e.g., FBR in Pakistan)?

■ *May require an Invoices/Tax table and specific reporting queries in SQL Server.*

Q26. Do you need to store return/refund records for legal or accounting purposes?

■ *Affects schema design for an Orders and Returns table with audit trails.*